Initial Project Planning

SimpliCI | CSE4104-7D-T01 | CSE 4104 Software Development III | Week 02 Submission

1. Scope Layers and Cut Line

Layer	Content	Target week	If behind schedule
Core MVP	F1 to F8: auth, input, 4 modes, language and level, domain presets, cited chat, history and feedback, admin dashboard	Working by Week 09	Never dropped
Differentiators	W1 evidence verification, W2 split-view, W3 Smart Alerts, W4 hover glossary, W6 export and share, W7 privacy, W8 AI analytics	W1 in Week 08, rest in Weeks 09 to 10	Drop W2 highlighting first, keep simple page jump
Could-have	W5 read aloud, stretch S1 (pgvector RAG) or S2 (quiz)	Only after Week 09 checkpoint	Dropped first

2. Team Roles and Ownership

Member	Role	Owns
Utsho Kumar Dey	Team Leader, AI Integration Lead	Architecture, prompts and JSON schemas, Gemini pipeline, evidence verification, GitHub setup and reviews, deployment, weekly reports
Md. Adnun Ahemed Methun	Backend Developer	Express APIs, JWT and bcrypt auth, role checks, upload handling, validation, rate limiting, API tests in Postman
Madhurjoo Dam	Frontend Developer	Figma wireframes, React pages, split-view reader, chat UI, loading and error states, responsive design, UI tests
Mst Amena Akter Akhi	Database Developer	ER diagram, Supabase PostgreSQL schema and storage buckets, analytics queries, dummy test documents and test question set

Every member also reviews at least one other member's pull request each week, so that everyone understands more than their own module for the viva.

3. Weekly Roadmap (Week 02 to Week 14)

Week	Handbook deliverable	SimpliCI plan	Lead
2	Proposal, team info, repo link, planning	Finalise scope layers. Create GitHub repository with README and folder structure. Add all collaborators.	Utsho
3	SRS and initial diagrams	Functional and non-functional requirements, User and Admin roles, use cases. Re-verify competitor table.	Utsho, Madhurjoo
4	Architecture, ER diagram, API structure	Architecture diagram, ER diagram, API list, prompt and JSON schema draft, use case diagram.	Utsho, Amena, Methun
5	Figma designs, UI screenshots, roadmap	Wireframes: login, dashboard, upload, result view, split view, chat, history, admin.	Madhurjoo
6	Backend code, API list, DB structure	Auth, database tables, file upload and storage, document and chat APIs.	Methun, Amena
7	Frontend and connected APIs	Auth pages, dashboard, upload, result and chat screens, API integration, responsive layout.	Madhurjoo, Methun
8	AI workflow explanation and source code	Page-wise extraction, vision for scans, domain presets, Gemini JSON pipeline, grounded chat, evidence verification (W1).	Utsho
9	Nearly complete project, feature checklist	Feedback, history, admin dashboard, Smart Alerts, glossary, export, privacy controls. MVP feature freeze and stretch decision.	All
10	Testing report, bug fix report	Postman API tests, UI tests, AI test set of 30 documents, quality report, bug fixes.	Methun, Madhurjoo, Utsho, Amena

Week	Handbook deliverable	SimpliCI plan	Lead
11	Live deployment link	Vercel, Render and Supabase setup, environment variables, warm-up plan for free-tier sleep.	Utsho, Methun
12	Final report, user manual, API docs	Each member writes the sections for their own module. Leader merges and edits.	All
13	Presentation slides	Slides, demo script, backup screenshots and demo video.	All
14	Final presentation and viva	Rehearsal where every member explains their own module and the AI workflow.	All

4. Milestones and Checkpoints

Milestone	Week	Done when
M1 Design freeze	4	Architecture, ER diagram and API list are approved by the team.
M2 Core AI pipeline	8	Upload to simplified JSON to cited chat answer works locally for all three domains.
M3 MVP freeze and stretch decision	9	F1 to F8 work together. Team decides which differentiators and which stretch goal remain.
M4 Live deployment	11	Public frontend and backend links work end to end with a test account.
M5 Demo rehearsal	13	Full demo runs without errors twice, with backup video ready.

5. Risks and Mitigation

Risk	Mitigation
Free-tier limits on Gemini and hosting (rate limits, file size, sleeping server)	File size and page limits, caching of results, retry with delay, warm up the Render service before demos.
Hallucinated answers	Answer-only-from-document prompt, evidence quote check in the backend (W1), "not found" response, thumbs feedback.
Poor Bangla OCR on scanned files	Test sample scans in Week 06 to 08, fall back to pasted text, show a clear quality warning.
Sensitive data	Dummy documents only during development, private storage, delete buttons, visible disclaimer.
Scope creep	Scope layers and cut line in section 1. New features only after the Week 09 checkpoint.
Uneven contribution or inactive member	Weekly task board, weekly pull requests per member, peer review of each other's code.
Model or API changes by the provider	Model name in an environment variable, prompts and schemas versioned in the repository.
Last-minute work	Weekly commits and submissions that show real progress, as the handbook requires.

6. GitHub Workflow

Item	Plan
Repository name	cse4104-7d-t01-simplici (format: cse4104-section-team-projectname)
Repository link	https://github.com/UTSHO2628/cse4104-7d-t01-simplici
Collaborators	All four members added as collaborators.
Branches	main (stable, demo-ready), development (integration), frontend, backend, ai-integration
Rules	No direct push to main. Work in feature branches, merge by pull request, at least one reviewer.

Item	Plan
Commits	Meaningful messages such as "Added JWT authentication" or "Implemented AI chatbot integration". No "update" or "final".
Frequency	Every member commits at least weekly. No large final-week uploads.
Secrets	API keys and database credentials only in .env files, never committed. A .env.example file is kept in the repository.
Task tracking	GitHub Projects board linked to the repository. Weekly team meeting and a weekly report by the leader.

7. Minimum Project Requirements Checklist

Requirement	Planned in SimpliCI	Status
Frontend	React (Vite) with Tailwind CSS	Planned, Weeks 05 and 07
Backend	Node.js and Express REST API	Planned, Week 06
Database	PostgreSQL on Supabase	Planned, Weeks 04 and 06
Authentication	JWT, bcrypt, User and Admin roles	Planned, Week 06
AI integration	Gemini structured simplification, grounded chat, evidence verification, vision for scans	Planned, Week 08
Deployment	Vercel, Render, Supabase	Planned, Week 11
Documentation	SRS, design, final report, user manual, API docs	Weeks 03, 04, 12
GitHub repository	cse4104-7d-t01-simplici	Created in Week 02
Testing	Postman, UI tests, AI test set	Planned, Week 10